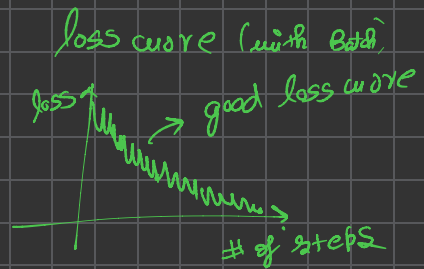
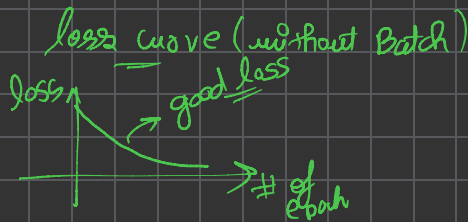


Loss wave in Neural Networks



loss wave:-

- 1) If loss wave is without batch, then will see, a lot smoother graph.
- 2) If loss wave is with batch, then will see more noise in the set.
- 3) In loss wave with batch, each step will be noisy.
- 4) If trained with batches, more steps are reqd.
- 5) If not trained with batches, less steps are reqd.
- 6) If trained with batches, then approx. grad. is calculated.
- 7) If not trained with batches, then exact grad. is calculated.

How to select a batch size?

- (A) It is not a magic some hyperparameter.
- (B) Batch general $\rightarrow 32/64/128$. (AlexNet:- 256)
- (C) whatever fits in your GPU memory, take that batch size.

Mini Batch SGD

1) In batches, we update the parameters

$$w_t = w_t - (\alpha \cdot \text{grad}_t)$$